

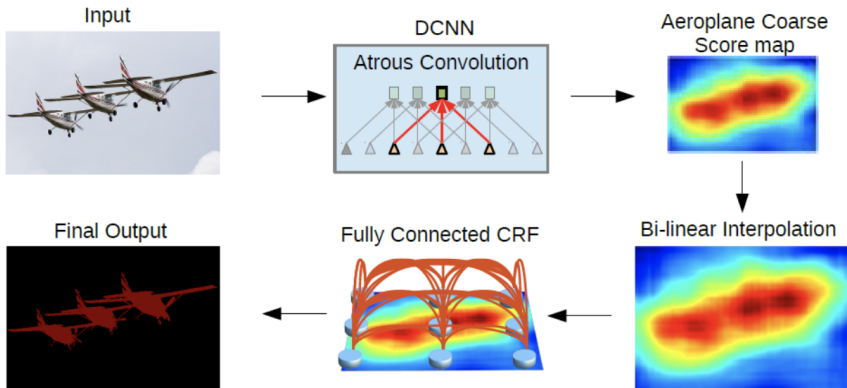
Introduction to neural networks

Laboratory of robotics and control systems
ITT PROJECT MANAGEMENT SERVICES L.L.C

Plan

1. Semantic segmentation
2. Losses and metrics
3. Interactive segmentation
4. Instance segmentation
5. Panoptic Segmentation

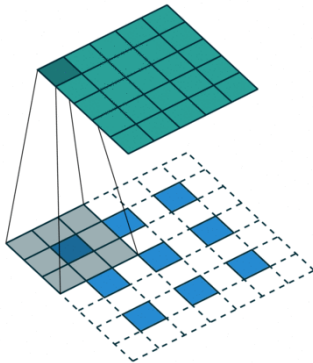
DeepLab (2016)



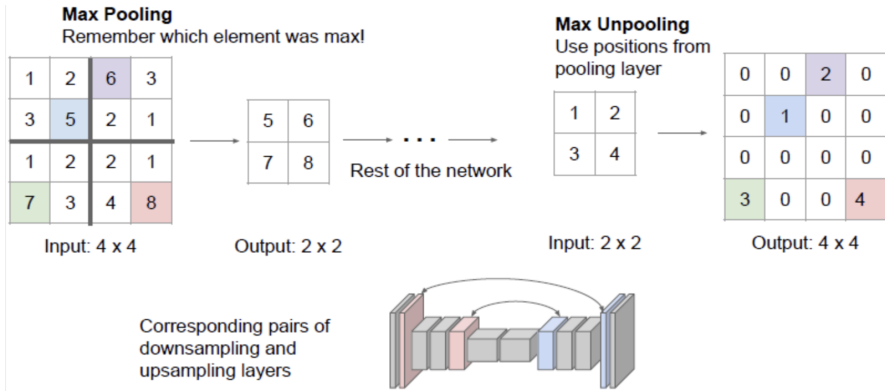
- Atrous convolution = Dilated convolution
- CRF = Conditional random field

Transposed convolution

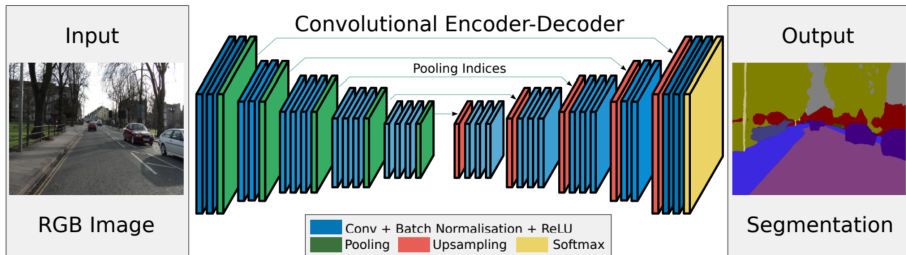
- Essentially a trainable upscaling matrix
- Can be initialized with a bilinear interpolation matrix



Max Unpooling



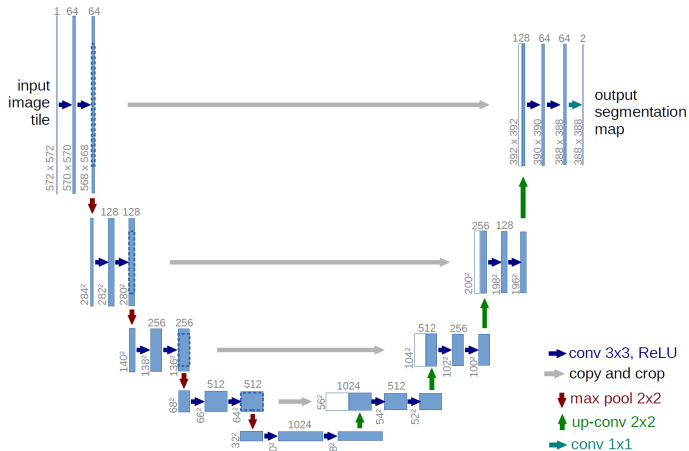
Segnet: encoder-decoder architecture (2015)



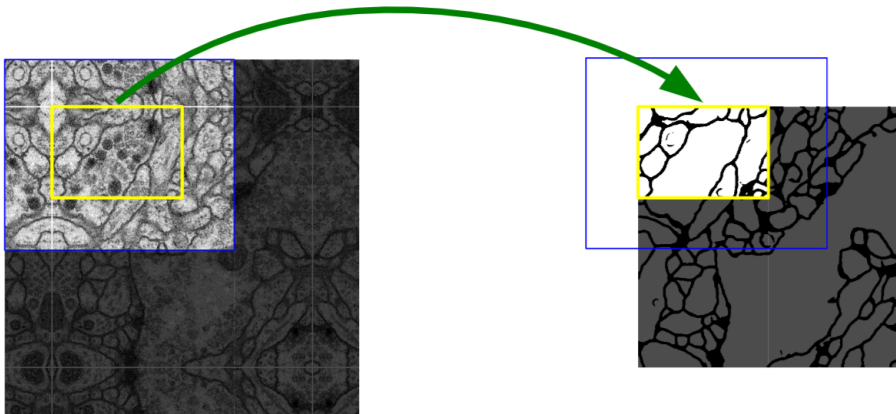
- Using pooling indices from encoder in decoder

U-Net: adding skip connections (2015)

- Input and output shapes don't match
- Skip connections help to restore masks
- Perform sum in feature fusion

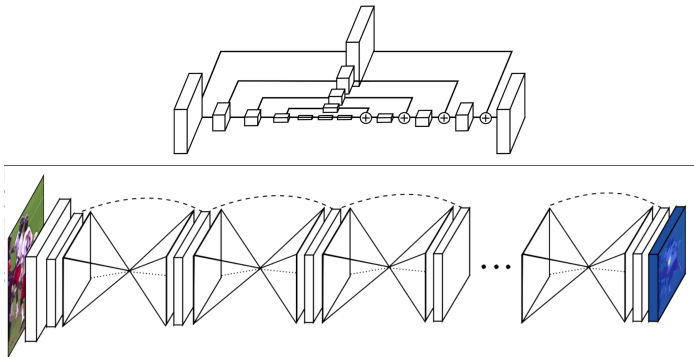


Predicting on images of arbitrary size

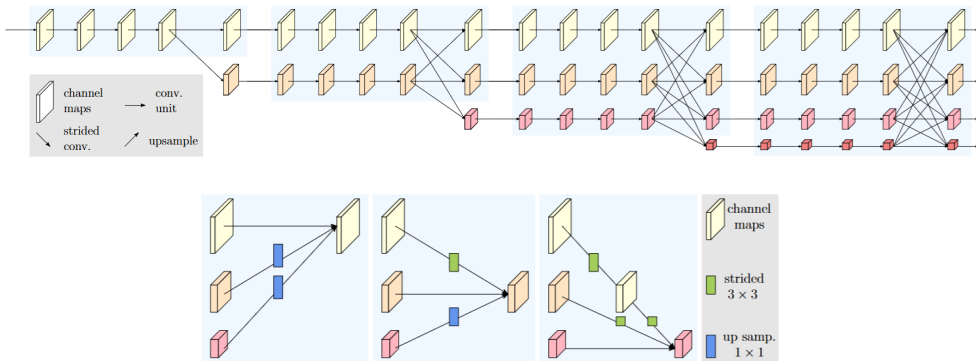


Hourglass Networks (2016)

- Using conv in skip connections
- Stack multiple U-Nets
- Using sum in features fusion



HRNet (2019)



- Every arrow in con layer
- Features are processed in parallel
- Output is the last yellow feature map

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Cross-entropy loss

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_i^c \log(\hat{y}_i^c)$$

Explanation:

- N : Number of pixels in the image.
- C : Number of classes.
- y_i^c : Ground truth label for pixel i and class c (one-hot encoded).
- $\hat{y}_i^c$: Predicted probability for pixel i and class c .

Features:

- Widely used in segmentation tasks.
- Penalizes incorrect class probabilities at each pixel.
- Effective for balanced datasets with sufficient training samples per class.

Weighted Cross-Entropy Loss

$$\mathcal{L}_{WCE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C w_c y_i^c \log(\hat{y}_i^c)$$

Explanation:

- w_c : Weight assigned to class c to handle class imbalance.

Features:

- Adjusts the importance of different classes, making it suitable for imbalanced datasets.
- Helps mitigate underrepresentation of rare classes.

Dice Loss

$$\mathcal{L}_{Dice} = 1 - \frac{2 \sum_{i=1}^N \sum_{c=1}^C y_i^c \hat{y}_i^c}{\sum_{i=1}^N \sum_{c=1}^C y_i^c + \sum_{i=1}^N \sum_{c=1}^C \hat{y}_i^c}$$

Explanation:

- Measures overlap between predicted and ground truth masks.

Features:

- Particularly effective for tasks with class imbalance.
- Focuses on regions where predictions and ground truths overlap, improving segmentation quality for small objects.

Focal Loss

$$\mathcal{L}_{Focal} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C (1 - \hat{y}_i^c)^\gamma y_i^c \log(\hat{y}_i^c)$$

Explanation:

- γ : Focusing parameter that controls the down-weighting of well-classified examples.

Features:

- Reduces the impact of easy-to-classify pixels.
- Improves performance on hard-to-classify regions and rare classes.

Tversky Loss

$$\mathcal{L}_{Tversky} = 1 - \frac{\sum_{i=1}^N \sum_{c=1}^C y_i^c \hat{y}_i^c}{\sum_{i=1}^N \sum_{c=1}^C (y_i^c \hat{y}_i^c + \alpha y_i^c (1 - \hat{y}_i^c) + \beta (1 - y_i^c) \hat{y}_i^c)}$$

Explanation:

- α and β : Control the penalties for false positives and false negatives.

Features:

- Generalizes Dice loss.
- Useful for highly imbalanced datasets where false positives and false negatives need different emphasis.

Combo Loss (Dice + Cross-Entropy)

$$\mathcal{L}_{Combo} = \lambda \mathcal{L}_{CE} + (1 - \lambda) \mathcal{L}_{Dice}$$

Explanation:

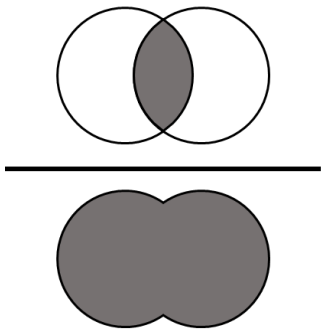
- Combines the strengths of cross-entropy and Dice loss.
- λ : Balancing factor between the two terms.

Features:

- Balances pixel-wise and region-wise metrics for robust performance.
- Suitable for datasets with moderate class imbalance.

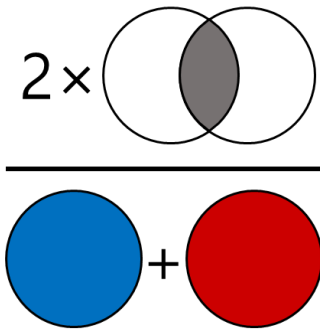
Segmentation metrics

IoU



$$IoU = \frac{\sum_{i=1}^N y_i \hat{y}_i}{\sum_{i=1}^N (y_i + \hat{y}_i - y_i \hat{y}_i)}$$

Dice Coefficient



$$Dice = \frac{2 \sum_{i=1}^N y_i \hat{y}_i}{\sum_{i=1}^N y_i + \sum_{i=1}^N \hat{y}_i}$$

Other segmentation metrics

- Precision = $\frac{TP}{TP+FP}$
- Recall = $\frac{TP}{TP+FN}$
- Component-wise precision and recall

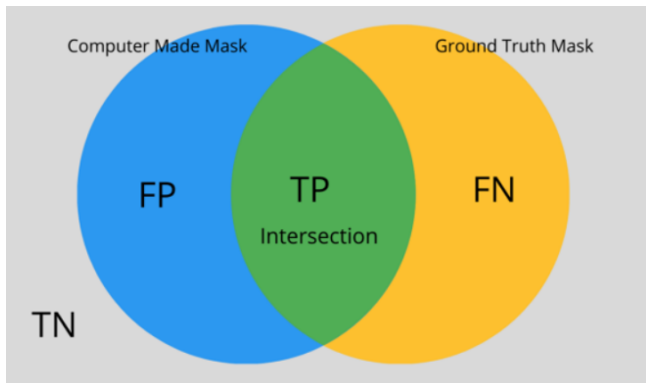


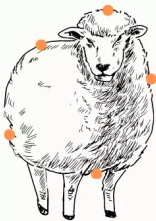
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Interaction UI

Interactive Image Segmentation

Points



Bounding Box



Scribbles



Evaluation

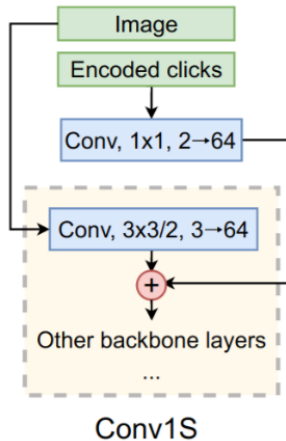
Datasets:

- GrabCut - 100 images
- DAVIS - 345 images
- Berkeley - 50 images
- SBD - 2857 images

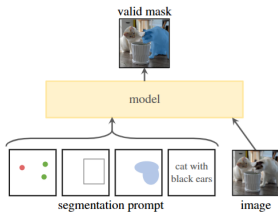
Metrics: Number of clicks to reach $\text{IoU} = 0.9$

RITM (2021)

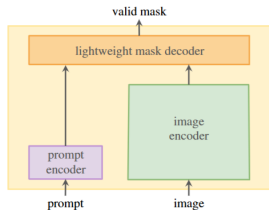
- Clicks encoding with small circles
- Iterative training
- Using previously predicted mask as input



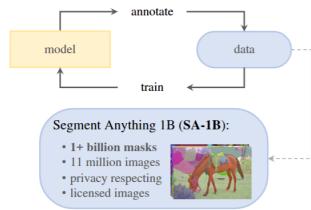
Segment anything (SAM) (2023)



(a) **Task:** promptable segmentation

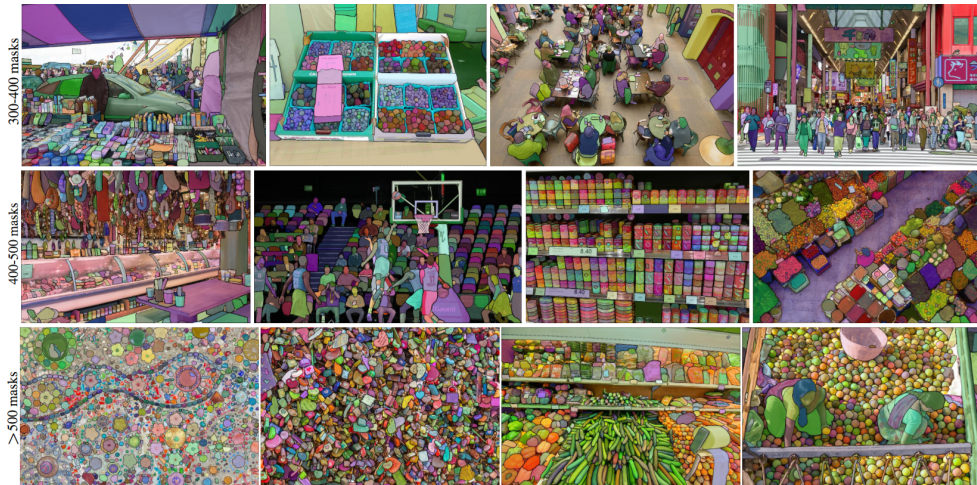


(b) **Model:** Segment Anything Model (SAM)



(c) **Data:** data engine (top) & dataset (bottom)

Dataset quality



Fast inference

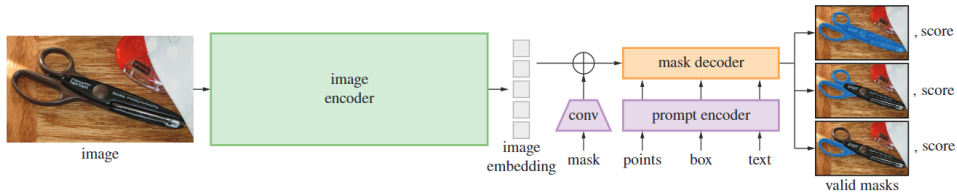


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Mask R-CNN (2017)

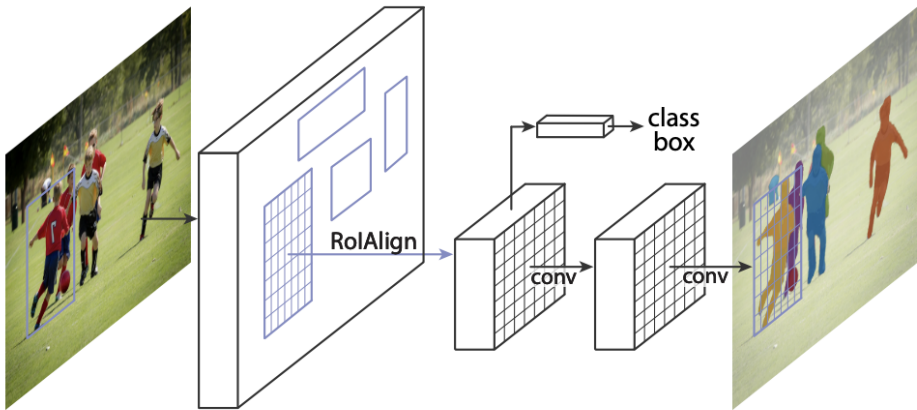
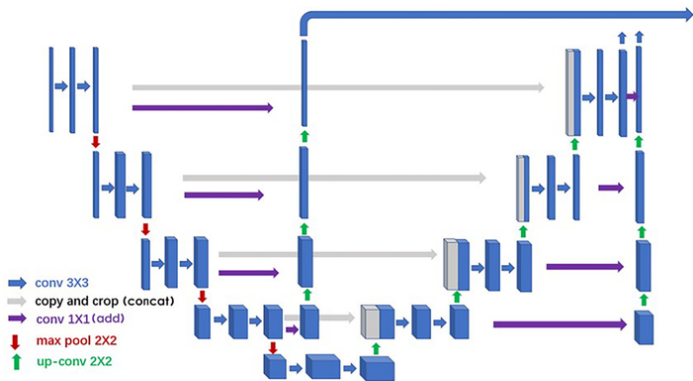


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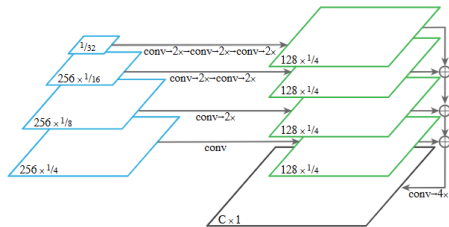
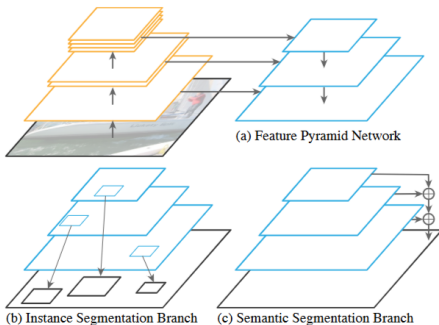
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Feature pyramid

- Captures multi-scale features, essential for understanding contexts of various sizes

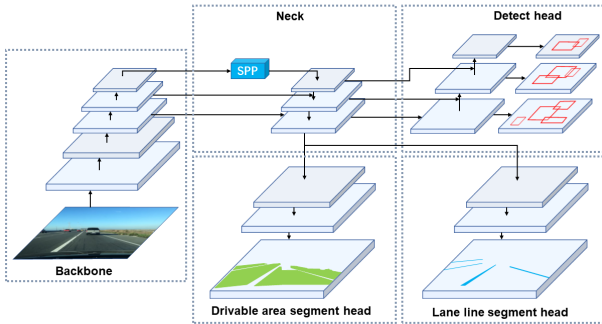


Panoptic Feature Pyramid Networks (2019)



YOLOP (2022)

- Detection Head: Similar to YOLOv5, uses anchors
- Drivable Area Head and Lane Line Head: same architecture, solve binary segmentation
- Uses a shared encoder to extract common features for all three tasks.





Thank you for your time

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